Exact Exchange in Orbital-Free Density Functional Theory

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Orbital-free density functional theory (OF-DFT) offers near-linear scaling and thus a lower computational cost compared to Kohn–Sham density functional theory (KS-DFT). However, a significant challenge has been the lack of an orbital-free expression for the exact exchange energy, which is explicitly orbital-dependent in the KS-DFT framework. In this work, we revisit the Levy–Perdew–Sahni formalism for the square root of the electron density, starting from a reference system of non-interacting boson-like fermions. This perspective preserves the conventional form of the theory while analytically demonstrating that the Fermi–Amaldi (FA) functional is the exact orbital-free exchange. We numerically investigate the effects of its inclusion. This advancement facilitates the development of functionals beyond the generalized gradient approximation (GGA) in OF-DFT, leveraging the accumulated experience from KS-DFT.

I. INTRODUCTION

The most popular practical method in density functional theory, Kohn–Sham DFT, is exact in principle and offers a favorable balance of accuracy and computational cost [1–3]. In KS-DFT, a set of single-electron equations

$$\left\{-\frac{1}{2}\nabla^2 + v_{\mathrm{KS}}([\rho];\mathbf{r})\right\}\varphi_i(\mathbf{r}) = \varepsilon_i\varphi_i(\mathbf{r}) \tag{1}$$

is solved to find KS orbitals $\{\varphi_i(\mathbf{r})\}$ and corresponding orbital energies $\{\varepsilon_i\}$. The KS potential is constructed to ensure that the density $\rho(\mathbf{r})$ of N non-interacting electrons

$$\rho(\mathbf{r}) = \sum_{i=i}^{N} |\varphi_i(\mathbf{r})|^2 \qquad \langle\varphi_i|\varphi_j\rangle = \delta_{ij} \tag{2}$$

is identical to the true density $\rho_0(\mathbf{r})$ of the interacting electrons. The implementation of KS-DFT scales nearly as $\mathcal{O}(N^3)$ limiting the applicability of KS-DFT to large systems, such as periodic structures and proteins [4].

An alternative, the orbital-free Levy–Perdew–Sahni (LPS) scheme, is emerging as a more universal choice for chemical modeling due to its near-linear scaling and reliance on pure density functionals [4–6]. Within the LPS framework, the variational minimization yields a single Euler–Lagrange equation for the square root of the electron density, effectively treating the system as N non-interacting bosons

$$\left\{-\frac{1}{2}\nabla^2 + v_{\mathrm{LPS}}([n];\mathbf{r})\right\}\sqrt{n(\mathbf{r})} = \mu\sqrt{n(\mathbf{r})}, \tag{3}$$

where μ represents the chemical potential, corresponding to the negative of the ionization energy [5]. The multiplicative LPS potential is similar to the KS potential and is constructed to make the non-interacting density of bosons equal to the physical electron density $\rho_0(\mathbf{r})$.

The widespread adoption of the LPS scheme is limited by the need for an approximate kinetic energy functional and an orbital-free exact exchange functional. Significant progress has been made in developing orbital-free kinetic energy approximations with a range of local, semi-local, and non-local functionals now available. This development is currently topped by the rapidly growing application of artificial intelligence, leading to machine-learned approximations. As for the exchange-correlation functional approximations, it seems natural to adopt them from KS-DFT, since it has been the primary focus of the field for the last decades. However, this massive experience cannot be always applied directly to the LPS scheme. The reason is that many of the most successful approximations in KS-DFT rely heavily on the exact exchange functional, which is explicitly orbital-dependent and has no known form as a pure density functional. Consequently, LPS is currently limited to GGA and meta-GGA exchange-correlation functionals [4, 6, 7].

To enable these advancements, a robust formulation of orbital-free exact exchange is the essential missing piece. In this paper, we reinterpret the original LPS scheme to formally derive the definition of orbital-free exact exchange. This finding paves the way for transferring concepts from KS-DFT functional development to the orbital-free LPS framework, particularly for higher rungs of the DFT Jacob's ladder, such as hybrid and long-range-corrected functionals [8, 9].

II. DERIVATION

Consider a fictitious system of non-interacting boson-like fermions with spin degeneracy $M = 2s + 1$ equal to the particle number. The Slater determinant for this system consists of N orthonormal spin-orbitals $\{\psi_i(\mathbf{x})\}$ each a product of a common spatial function $\phi(\mathbf{r})$ and orthogonal spin function $\{\sigma_i(\omega)\}$. Since each particle occupies a distinct spin state, the Pauli exclusion principle allows them to share the same spatial orbital. Consequently, the antisymmetry is confined to the spin component, and the wavefunction factors as

$$\Psi(\mathbf{x}_1,\ldots,\mathbf{x}_N) = \frac{\Phi_{\mathrm{HP}}(\mathbf{r}_1,\ldots,\mathbf{r}_N)}{\sqrt{N!}}\det\{\sigma_i(\omega_j)\}, \tag{4}$$

where the spatial part Φ_{HP} is the Hartree product [10]. The corresponding density for the system of N non-interacting boson-like fermions is given by

$$n(\mathbf{r}) = |\phi(\mathbf{r})|^2 \qquad \langle\phi|\phi\rangle = N. \tag{5}$$

The non-interacting kinetic energy, $T_{\mathrm{s}}[n]$, is obtained by evaluating the expectation value of the kinetic energy operator using the wavefunction Ψ defined in Eq. (4). This yields the von Weizsäcker kinetic energy functional [11]

$$T_{\mathrm{s}}[n] \equiv T_{\mathrm{W}}[n] = \int \sqrt{n(\mathbf{r})}\left(-\frac{1}{2}\nabla^2\right)\sqrt{n(\mathbf{r})}d\mathbf{r}. \tag{6}$$

The total *electronic* energy, formulated using the non-interacting boson-like kinetic energy (6), is given by

$$E[n] = T_{\mathrm{W}}[n] + V_{\mathrm{ext}}[n] + J[n] + G[n], \tag{7}$$

where $J[n]$ represents the classical Coulomb self-repulsion energy

$$J[n] = \frac{1}{2}\int\int \frac{n(\mathbf{r})n(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}d\mathbf{r}d\mathbf{r}', \tag{8}$$

and the functional

$$G[n] = T_{\mathrm{P}}[n] + E_{\mathrm{xc}}[n] \tag{9}$$

combines both the exchange-correlation energy, $E_{\mathrm{xc}}[n]$, and the Pauli energy, $T_{\mathrm{P}}[n]$. Physically, the exchange-correlation functional accounts for non-classical electron-electron interactions, while the Pauli energy functional corrects the kinetic energy (6) to recover the electronic spin-statistics. Consequently, the former requires an exchange-correlation approximation, while the latter needs a kinetic energy density functional (KEDF) approximation. The orbital-free effective potential of Eq. (3) corresponding to the energy functional (7) is given by the sum of the KS potential of Eq. (1) and the Pauli potential

$$v_{\mathrm{LPS}}([n];\mathbf{r}) = v_{\mathrm{KS}}([n];\mathbf{r}) + \frac{\delta T_{\mathrm{P}}[n]}{\delta n(\mathbf{r})}, \tag{10}$$

$$v_{\mathrm{KS}}([n];\mathbf{r}) = v_{\mathrm{ext}}(\mathbf{r}) + \int \frac{n(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}d\mathbf{r}' + \frac{\delta E_{\mathrm{xc}}[n]}{\delta n(\mathbf{r})}. \tag{11}$$

Thus, we have recovered the conventional orbital-free energy functional, equivalent to the one derived by Levy, Perdew, and Sahni [5]. The difference lies in the specific choice of the wavefunction, Eq. (4) and the associated particle spin statistics. This approach effectively replaces the fictitious Kohn–Sham reference system of non-interacting electrons with a reference system of non-interacting boson-like fermions.

The exact exchange functional in KS-DFT is defined as the expectation value of the two-electron operator, $\hat{V}_{\mathrm{ee}}$, minus the classical Coulomb repulsion energy $J[\rho]$ (8)

$$E_{\mathrm{x}}^{\mathrm{exact}}[\rho] = \langle\Psi_{\min}|\hat{V}_{\mathrm{ee}}|\Psi_{\min}\rangle - J[\rho], \tag{12}$$

$$\hat{V}_{\mathrm{ee}} = \sum_{i<j}^{N} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}, \tag{13}$$

where $\Psi_{\min}$ is of the form of the wave function (4) minimizing the energy functional (7). Since the two-particle operator (13) is spin-independent, the exact exchange definition (12) applies directly to a wave function describing particles of arbitrary type and spin, e.g., boson-like fermions. Substitution of Eq. (4) to Eq. (12) yields the Fermi–Amaldi functional

$$E_{\mathrm{x}}^{\mathrm{exact}}[n] \equiv E_{\mathrm{x}}^{\mathrm{FA}}[n] = -\frac{1}{N}J[n]. \tag{14}$$

This aligns with our previous finding that the Fermi–Amaldi functional is the exact exchange energy for any system of particles sharing a single spatial orbital, regardless of their spin and type [12, 13].

III. COMPUTATIONAL STUDY

In this section, we numerically solve Eq. (3) using various approximations for the $G[n]$ functional (9). Our objective is to assess the impact of the FA functional on atomic energies, dissociation curves, and electron densities. The FA potential is given by

$$v_{\mathrm{x}}^{\mathrm{FA}}([n];\mathbf{r}) = -\frac{1}{N}\int \frac{n(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}d\mathbf{r}, \tag{15}$$

where N is treated as a constant parameter [14–20]. The FA functional is tested against the local LDA [21, 22] and the semi-local (GGA) PBE [23, 24] exchange functionals. The exchange functionals are paired with local, GGA, and meta-GGA KEDFs. Non-local and machine-learned KEDFs are not included into this study due to the non-trivial implementation within GTO basis set.

A. Implementation Details

To implement the LPS scheme of Eq. (3), we adapted the open-source Psi4NumPy code [25, 26]. In the adapted scheme, the density matrix is constructed exclusively from the lowest-energy eigenfunction normalized to N [5]. A Core Hamiltonian initial guess was employed, and all molecular integrals were computed analytically using a Gaussian basis set [10]. The approximate functional $G[n]$ (9) was evaluated numerically using a Lebedev–Treutler $(999, 974)$ quadrature grid. In difficult convergence cases, the electron density calculated at smaller grids, e.g., $(350, 74)$, was used as an initial guess for the larger grid calculations.

TABLE I. Comparison of the Fermi–Amaldi exchange to pure LDA and PBE exchange functionals across the range of TFλW models varying by the value of the Weizsäcker functional fraction λ. MAEs and rMAEs are given with respect to UHF/UGBS SCF atomic (H–Ne) energies

KEDF	TF$\frac{1}{9}$W	TF$\frac{1}{6}$W	TF0.186W	TF$\frac{1}{5}$W	TF$\frac{1}{3}$W	TFW
			Mean Absolute Error (a.u.)			
LDA	4.38	1.58	0.76	0.24	4.23	15.74
PBE	4.95	2.11	1.26	0.69	3.78	15.44
FA	2.91	0.23	0.65	1.19	5.48	16.67
			Relative Mean Absolute Error (%)			
LDA	9.67	3.72	2.09	1.72	12.24	40.24
PBE	12.54	5.10	3.03	1.88	10.07	38.86
FA	8.00	1.14	1.66	2.71	13.41	41.00

To obtain the self-consistent solutions, we used a combination of the direct inversion of the iterative subspace (DIIS) procedure [27] with Hartree damping [28] and Level-Shifting [29] techniques. The maximum length of the DIIS vector was varied from 3 to 9 depending on the system. All eigenvalues except the lowest one were shifted up by $0.1 - 0.9$ at the beginning of the SCF cycle returning it back to 0.0 closer to the convergence. The damping factor was varied in the range $0.1 - 0.999$. The convergence criteria for both the total energy and RMSD of the density matrix was set to 1.0E-5 Hartree and unitless change, respectively.

To validate our implementation, we reproduced the reference H–Ne results of Chan *et al.* [30] and Karasiev *et al.* [31]. For open-shell atoms, this required adapting our code to match the spin-restricted approach used in those studies. In contrast, the current work utilizes the spin-polarized formalism: the α and β density matrices are generated from the lowest-energy eigenfunctions normalized to N_α and N_β, respectively.

B. Self-Consistent Atomic Energies

Self-consistent calculations were performed using the Thomas–Fermi-λ-Weizsäcker (TFλW) KEDF [30–33]. Thus, Eq. (3) was solved for functionals (7) and (9) with

$$T_{\mathrm{P}}^{\mathrm{TF}\lambda\mathrm{W}}[n] = T_{\mathrm{TF}} + (\lambda - 1)T_{\mathrm{W}}, \tag{16}$$

where

$$T_{\mathrm{TF}}[n] = C_{\mathrm{F}} \int n^{5/3}(\mathbf{r})d\mathbf{r} \qquad C_{\mathrm{F}} = \frac{3}{10}\left(3\pi^2\right)^{2/3}. \tag{17}$$

Table I presents the mean absolute errors MAE $= \left(\sum_a \Delta E_a\right)/N_A$, where $\Delta E_a = |E_a^{\mathrm{LPS}} - E_a^{\mathrm{UHF}}|$, across $N_A = 10$ atoms, in a.u., and the relative mean absolute errors rMAE $= \left(-\sum_a \Delta E_a / E_a^{\mathrm{UHF}}\right)/N_A \times 100\%$ expressed as a percentage of the UHF/UGBS energy.

Specific λ values were taken from the literature (see [30, 31] and references therein). Despite the known deficiencies [4, 34], the TFλW model was selected because it is one of the few KEDFs for which self-consistent convergence, though difficult, is manageable [30, 31]. To maintain the spherical symmetry of the atoms and assist the convergence, a UGBS basis set was contracted to 31 s-type Gaussians, $\exp(-\alpha r^2)$, with exponents α ranging from $1.14 \times 10^7 \ldots 2.00 \times 10^{-2}$.

In alignment with other authors [30, 31], Table I shows that the optimal von Weizsäcker mixing parameter for the PBE and LDA pure exchange functionals is $\lambda = 1/5$. As for the FA functional, the optimal value is $\lambda = 1/6$, with the second best being $\lambda = 0.186$. This result indicates that the FA functional can outperform both LDA and PBE exchange functionals when paired with an appropriate KEDF.

C. Atomic Densities

Self-consistent radial distribution functions for the Ne atom are presented in Fig. 1. The optimal TFλW kinetic functionals were employed for each exchange functional. All calculations used the modified UGBS basis set described in the previous section.

The TF$\frac{1}{6}$W FA density decays more rapidly than either the TF$\frac{1}{5}$W LDA or TF$\frac{1}{5}$W PBE densities. As expected, the choice of exchange functional does not fundamentally alter the density structure, i.e., the FA exchange does not recover atomic shell structure, which remains absent due to the monotonic nature of the TFλW kinetic model. The more rapid asymptotic decay observed in the FA density is consistent with established results from statistical atomic theory [35].

D. Dissociation Profiles

Figure II shows the H_2 dissociation curves computed using the same models as in the previous section for density plotting. These calculations employed the UGBSP basis set, a modified UGBS containing 13 s- and 4 p-

TABLE II. Comparison of FA post-SCF energies with LDA and PBE exchange functionals for precomputed atomic densities (H–Ne). SCF densities were obtained at the UHF/UGBS level of theory. Mean absolute errors (MAE) and relative MAEs (rMAE) are reported relative to the UHF/UGBS self-consistent energies.

KEDF	TF	TF$\frac{1}{9}$W	TF$\frac{1}{6}$W	TF0.186W	TF$\frac{1}{5}$W	TF$\frac{1}{3}$W	Weizsäcker	TW
Mean Absolute Error (a.u.)								
LDA	3.47	0.37	2.29	2.95	3.44	8.04	9.47	0.52
PBE	3.97	0.21	1.78	2.45	2.93	7.53	9.93	0.07
FA	1.92	1.91	3.83	4.50	4.98	9.58	7.90	2.06
Relative Mean Absolute Error (%)								
LDA	6.78	3.02	7.86	9.54	10.76	22.36	13.17	2.97
PBE	9.09	0.93	5.41	7.09	8.31	19.91	13.06	0.60
FA	5.38	4.29	9.12	10.80	12.02	23.62	9.35	4.24

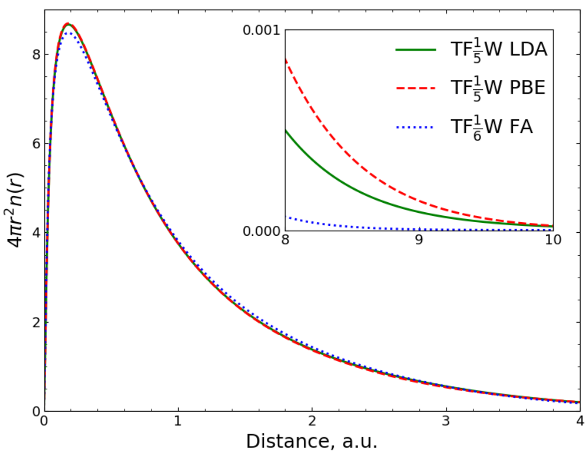

FIG. 1. Comparison of the Ne radial distribution function $4\pi r^2 n(\mathbf{r})$ calculated with the TF$\frac{1}{6}$W FA functional to TF$\frac{1}{5}$W LDA and TF$\frac{1}{5}$W PBE functionals using 31 s-type basis functions from the UGBS basis set.

functions ($\alpha_s \in [9.34\times10^2, 7.67\times10^{-2}]$ and $\alpha_p \in [5.76\times10^{-1}, 3.92\times10^{-2}]$) for each hydrogen. An RHF/UGBSP curve is included as the reference. Although spin-polarized functionals were used, our LPS implementation lacks a symmetry-breaking mechanism. As such, it converges to the spin-symmetric solution, making RHF the appropriate spin-symmetric reference rather than the broken-symmetry UHF solution.

As shown in Figure II, the FA functional exhibits better long-range behavior and a distinct potential well unlike both the LDA and PBE functionals. The FA functional also yields an equilibrium distance, r_e, significantly closer to the RHF reference, e.g., 2.00 a.u. against 3.00 and 3.25 a.u. for LDA and PBE, respectively. The FA binding energy estimate, $E(4\,\text{a.u.}) - E(r_e) = 0.0134$ a.u., is closer to the RHF reference (0.2191 a.u.) compared to 0.0028 and 0.0007 a.u. for LDA and PBE, respectively.

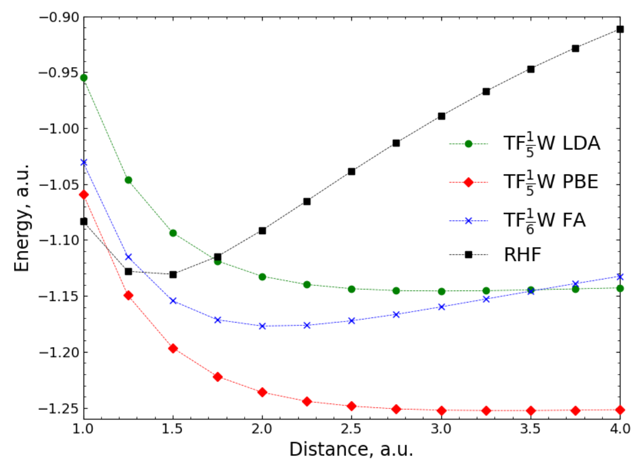

FIG. 2. Comparison of the TF$\frac{1}{6}$W FA functional to TF$\frac{1}{5}$W LDA and TF$\frac{1}{5}$W PBE functionals for the H_2 potential energy surface computed with the UGBSP basis set. Self-consistent calculations were performed only at the marked points. The connecting curves are interpolations.

E. Post-SCF Atomic Energies

The FA functional's promising energetic performance on one hand with the absence of shell structure in the self-consistent densities on another prompted us to evaluate its accuracy in post-SCF calculations. The set of self-consistent UHF/UGBS atomic densities for H–Ne was used to compute post-SCF atomic energies with the FA, LDA, and PBE exchange functionals.

The range of KEDFs employed in this section includes the TFλW functionals from Table I as well as the pure TF [36, 37], the pure Weizsäcker [11], and the Tran–Wesołowski (TW) [38] functionals. The latter was chosen because it shares the functional form with PBE [23, 24, 39, 40] and Becke86A [41] exchange functionals making it a good pair to the PBE exchange. The Pauli term for the TW KEDF is given by

$$T_{\mathrm{P}}^{\mathrm{TW}}[n] = C_{\mathrm{F}} \int n^{5/3}(\mathbf{r}) F_{\mathrm{TW}}(s) - T_{\mathrm{W}}[n], \tag{18}$$

where $s = |\nabla n|/(2k_{\mathrm{F}}n)$ is the reduced density gradient, $k_{\mathrm{F}} = (3\pi^2 n)^{1/3}$ is the Fermi wave vector. The $T_{\mathrm{W}}[n]$ is subtracted to isolate the Pauli contribution according to Eqs. (7) and (9), and $F_{\mathrm{TW}}(s)$ is the Tran–Wesołowski enhancement factor

$$F_{\mathrm{TW}}(s) = 1 + \kappa - \frac{\kappa}{1 + \frac{\mu}{\kappa} s^2}, \tag{19}$$

with empirical parameters $\kappa = 0.8209$ and $\mu = 0.2335$ fitted to He and Ne atoms [38].

Table II summarizes the mean absolute errors (MAE) and relative mean absolute errors (rMAE) for each exchange-KEDF combination. Overall, the performance of the approximate exchange functionals strongly depends on the choice of the KEDF. The FA functional yields consistently higher errors than both LDA and PBE for all tested KEDFs except the pure TF functional. When paired with the TFλW model, all exchange functionals achieve their lowest MAE and rMAE at $\lambda = 1/9$ which corresponds to the second-order gradient expansion of the TF functional. As expected, the PBE functional performs best when paired with the TW KEDF.

IV. CONCLUSION

In this work, we have reinterpreted the Levy–Perdew–Sahni (LPS) scheme to demonstrate that the Fermi–Amaldi functional is the exact exchange energy within the orbital-free density functional theory. By utilizing a reference system of non-interacting boson-like fermions, where all particles share a single spatial orbital, we showed that the conventional LPS energy functional [5] is fully recovered and the exact exchange is easily defined.

The Fermi–Amaldi functional exhibits several highly desirable properties for an exchange functional. It is self-interaction free for one-orbital systems [12, 42], possesses the correct long-range asymptotic behavior [42], and exhibits the necessary derivative discontinuity at integer electron numbers N [14]. Furthermore, unlike the exact Kohn–Sham exchange potential, the Fermi–Amaldi potential is strictly multiplicative. Recently, we also extended the FA functional to fractional electron number [14]. Collectively, these features identify the FA functional as a promising building block for future developments in OF-DFT.

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